

Abstract I

template_autopoiesis is a combinatoric grammar that **deterministically generates whole runnable projects** — not files or snippets, but complete, independently testable child repositories with their own kernel source, tests, analysis entry point, and manuscript. A single integer seed plus a grammar of orthogonal slots (primitive domain, analytical track, section set, and three presentation/provenance slots) selects one child from a combinatoric product space of 360 nominal (45 content-distinct) configurations, via a SHA-256 digest of the seed and slot identity — with no random-number generator anywhere in the expansion path.

Abstract II

Project generators routinely claim completeness, determinism, and traceability without making any of the three independently checkable. This exemplar treats each claim as a structural property to verify rather than a rhetorical one to assert: `verify_child()` recomputes a tree hash from the files actually on disk and compares it against the recorded provenance, rather than trusting a value the same run wrote down; a honesty manifest inspects the live source AST to confirm every claim in this manuscript resolves to a real function in a real file; and a per-domain mutation gate checks that the acceptance tests reject a constant-success stub before trusting that they accept the real kernel. The same discipline governs this document itself — every number below is substituted at render time from a live measurement rather than hand-typed as a literal.

Abstract III

Across 5 heterogeneous primitive domains (- optimization - dynamics - statistics - signal - graph), 493 tests exercise both fixed ground-truth checks and Hypothesis-driven property invariants at 96.28% branch coverage, with an explicit negative control per domain distinguishing the real kernel from a deliberately-wrong one.

Generation pipeline

flowchart LR

```
G[Grammar\nconfig.yaml] --> E[Expand\nseed → Spec]  
E --> M[Materialize\nSpec → Child]  
M --> V[Verify\ntree-hash check]  
V --> S[Seal\nQR provenance]
```

Abstract IV

Grammar product space

```
%% noqa: docs-lint - {{...}} tokens are manuscript-variable
%% (src/manuscript_variables.py) injected at render time, n
flowchart TB
```

```
    A[primitive_domain: 5] --> EP[Effective product: 45 cel
```

```
    T[track: 3] --> EP
```

```
    S[section_set: 3] --> EP
```

```
    EP --> TP[Total product: 360 cells]
```

```
    R[3 reserved slots, 2 options each] --> TP
```

- ▶ **Domain count:** 5
- ▶ **Effective product size:** 45
- ▶ **Total product size:** 360
- ▶ **Reserved slots:** 3 (figure_profile, qr_profile, integrity_profile)
- ▶ **Grammar hash:** f84a8f9dbcb18e37
- ▶ **Tests:** 493 · **Coverage:** 96.28%